Chapter 5 Statistical estimation

XIE Wenxian

School of mathematics and statistics
Northwestern Polytechnical University

October 28, 2024

Contents

Contents

5.1 Introduction

- Some statistical methods to find estimators of population parameters
  - Point estimations of the unknown population parameters
  - Interval estimations of the unknown population parameters
- The properties of estimations

Contents

5.2 The methods of finding point estimators

Let $X_1, \ldots, X_n$ be independent and identically distributed (iid) random variables (in statistical language, a random sample) with a probability density function (pdf) or probability mass function (pmf) $f(x; \theta_1, ., \theta_l)$, where $\theta_1, ., \theta_l$ are the unknown population parameters (characteristics of interest)

- Point estimation
  - Estimate the value of each of the parameters
  - Assign an appropriate value for the parameters $\theta = (\theta_1, \ldots, \theta_l)$ based on observed sample data from the population
  - Determine statistics $g_i(X_1, \ldots, X_n), i = 1, \ldots, l$
- Criteria for choosing a desired point estimator
  - unbiased, sufficient, efficient
- Methods
  - The method of moments and the method of maximum likelihood

5.2 The methods of finding point estimators

- Estimators
  - Estimators of θ_i: $\hat{\theta}_i = g_i(X_1, \dots, X_n), i = 1, \dots, l$
  - An estimator of $\theta = (\theta_1, \dots, \theta_l)$: $\hat{\theta} = (\hat{\theta_1}, \dots, \hat{\theta_l})$ denoted by $\hat{\theta}(X_1, \dots, X_n)$
  - Estimators for the parameters: statistics
- Estimates
  - The values calculated from estimators using particular sample data values
  - Denoted by $\hat{\theta}(x_1, \dots, x_n)$
  - $x_1, \dots, x_n$ are the observed values of the random variables $X_1, \dots, X_n$

Example: in case of the normal distribution

- Estimators of $\theta_1 = \mu$ and $\theta_2 = \sigma^2$: $\bar{X}$ and S^2
- The corresponding estimates: $\bar{x}$ and s^2

5.2.1 The method of moments

- It is based on matching the sample moments with the corresponding population (distribution) moments
- It is founded on the assumption that sample moments should provide good estimates of the corresponding population moments
- The population moments are often functions of the population parameters
- The moment estimator is not unique
- The moment estimator is easy to compute

5.2.1 The method of moments

The method of moments procedure Suppose there are l parameters to be estimated

- (1) Find l population moments: the kth moment about the origin of a random variable X whenever it exists:

$$\mu_k^{'} = E[X^k] = h_k(\theta_1, \dots, \theta_l)$$

- (2) Find l sample moments: the corresponding kth sample moment

$$m_k^{'} = \frac{1}{n} \sum_{i=1}^{n} X_i^k$$

- (3) From the system of equations, $\mu_k^{'} = m_k^{'}, k = 1, 2, \dots, l$ solve for the parameter $\theta = (\theta_1, \dots, \theta_l)$ as a moment estimator $\hat{\theta}$

5.2.1 The method of moments

Example 5.2.1 Let $X_1, \dots, X_n$ be a random sample from a Bernoulli population with parameter p.

(a) Find the moment estimator for p

(b) Tossing a coin 10 times and equating heads to value 1 and tails to value 0, we obtained the following values:

$$0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0.$$

Obtain a moment estimate for p, the probability of success (head).

Example 5.2.1 Let $X_1, \dots, X_n$ be a random sample from a Bernoulli population with parameter p.

(a) Find the moment estimator for p

(b) Tossing a coin 10 times and equating heads to value 1 and tails to value 0, we obtained the following values:

$$0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0.$$

Obtain a moment estimate for p, the probability of success (head).

Solution

(a)

$$\mu_k' = E[X] = p$$

$$m_1' = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Then, the method of moments estimator for p is $\hat{p} = \frac{\sum_{i=1}^{n} X_i}{n}$.

(b) Using (a) with $Y = \sum_{i=1}^{n} X_i = 6$, we get the moment estimate of p as

$$\hat{p} = \frac{6}{10} = 0.6$$

We would use this value $\hat{p} = 0.6$ to answer any probabilistic questions for the given problem. For example, what is the probability of obtaining exactly 8 heads out of 10 tosses of this coin? This can be obtained by using the binomial formula with $\hat{p} = 0.6$; that is,

$$P(X = 8) = \binom{10}{8}(0.6)^8(0.4)^{10-8} = 0.1209324.$$

Example 5.2.2 Let $X_1, \ldots, X_n$ be a random sample from a gamma probability distribution with parameters α and β. Find moment estimators for the unknown parameters α and β.

5.2.1 The method of moments

Example 5.2.2 Let $X_1, \dots, X_n$ be a random sample from a gamma probability distribution with parameters α and β. Find moment estimators for the unknown parameters α and β.

Solution

For the gamma distribution (see Section 3.2.5),

$$E[X] = \alpha\beta \text{ and } E\left[X^2\right] = \alpha\beta^2 + \alpha^2\beta^2.$$

Because there are two parameters, we need to find the first two moment estimators. Equating sample moments to distribution (theoretical) moments, we have:

$$\frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X} = \alpha\beta, \text{ and } \frac{1}{n}\sum_{i=1}^{n} X_i^2 = \alpha\beta^2 + \alpha^2\beta^2.$$

5.2.1 The method of moments

Solving for α and β, we obtain the estimates as $\alpha = \frac{\overline{X}}{\beta}$ and $\beta = \left[\{(\frac{1}{n})\sum_{i=1}^{n} X_i^2 - \overline{X}^2\}/\overline{X}\right]$.
Therefore, the method of moments estimators for α and β are:

$$\hat{\alpha} = \frac{\overline{X}}{\hat{\beta}} \text{ and } \hat{\beta} = \frac{\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \overline{X}^2}{\overline{X}} = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n\overline{X}},$$

which implies that:

$$\hat{\alpha} = \frac{\overline{X}}{\hat{\beta}} = \frac{\overline{X}^2}{\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \overline{X}^2} = \frac{\overline{X}^2}{\sum_{i=1}^{n}(X_i - \overline{X})^2}$$

Thus, we can use these values in the gamma pdf to answer questions concerning the probabilistic behavior of the random variable X.

5.2.1 The method of moments

Example 5.2.3 Let the distribution of X be $N(\mu, \sigma^2)$.

(a) For a given sample of size n, use the method of moments to estimate μ and σ^2.

(b) The following data (rounded to the third decimal digit) were generated from a normal distribution with mean 2 and standard deviation of 1.5:

3.163	1.883	3.252	3.716	-0.049	-0.653	0.057	2.987
4.098	1.670	1.396	2.332	1.838	3.024	2.706	0.231
3.830	3.349	-0.230	1.496				

Obtain the method of moments estimates of the true mean and the true variance.

Example 5.2.3 Let the distribution of X be $N(\mu, \sigma^2)$.

(a) For a given sample of size n, use the method of moments to estimate μ and σ^2.

(b) The following data (rounded to the third decimal digit) were generated from a normal distribution with mean 2 and standard deviation of 1.5:

3.163	1.883	3.252	3.716	-0.049	-0.653	0.057	2.987
4.098	1.670	1.396	2.332	1.838	3.024	2.706	0.231
3.830	3.349	-0.230	1.496				

Obtain the method of moments estimates of the true mean and the true variance.

Solution

(a) For the normal distribution, $E(X) = \mu$, and because $Var(X) = E(X^2) - \mu^2$, we have the second moment as $E(X^2) = \sigma^2 + \mu^2$.

5.2.1 The method of moments

Equating sample moments to distribution moments we have:

$$\frac{1}{n}\sum_{i=1}^{n} X_i = \mu_1' = \mu \text{ and } \mu_2' = \frac{1}{n}\sum_{i=1}^{n} X_i^2 = \sigma^2 + \mu^2$$

Solving for μ and σ^2, we obtain the moment estimators as:

$$\hat{\mu} = \overline{X} \text{ and } \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \overline{X}^2 = \frac{1}{n}\sum_{i=1}^{n} \left(X_i - \overline{X}\right)^2$$

(b) Because we know that the estimator of the mean is $\hat{\mu} = \overline{X}$ and the estimator of the variance is $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \overline{X}^2$, from the data the estimates are $\hat{\mu} = 2.005$, and $\hat{\sigma}^2 = 6.12 - (2.005)^2 = 2.1$. Notice that the true mean is 2 and the true variance is 2.25, which we used to simulate the data.

Example 5.2.4 Let $X_1, \cdots, X_n$ be a random sample from a uniform distribution on the interval $[a, b]$. Obtain method of moment estimators for a and b.

Example 5.2.4 Let $X_1, \cdots, X_n$ be a random sample from a uniform distribution on the interval $[a, b]$. Obtain method of moment estimators for a and b.

Solution

The pdf of a uniform distribution is given by:

$$f(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a \le x \le b \\ 0, & \text{otherwise} \end{cases}$$

The first two population moments are:

$$\mu_1 = E(X) = \int_a^b \frac{x}{b-a}\, dx = \frac{a+b}{2}$$

and

$$\mu_2 = E(X^2) = \int_a^b \frac{x^2}{b-a}\, dx = \frac{a^2 + ab + b^2}{3}$$

5.2.1 The method of moments

The corresponding sample moments are:

$$\hat{\mu}_1 = \bar{X} \quad \text{and} \quad \hat{\mu}_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2$$

Equating the first two sample moments to the corresponding population moments, we have:

$$\hat{\mu}_1 = \frac{a+b}{2} \text{ and } \hat{\mu}_2 = \frac{a^2+ab+b^2}{3}$$

Solving for a and b, we obtain the moment estimators of a and b:

$$\hat{a} = \hat{\mu}_1 - \sqrt{3(\hat{\mu}_2 - \hat{\mu}_1^2)} \text{ and } \hat{b} = \hat{\mu}_1 + \sqrt{3(\hat{\mu}_2 - \hat{\mu}_1^2)}$$

Example 5.2.5 Let $X_1, \ldots, X_n \sim \text{Poisson}(\lambda)$. Show that both $\frac{1}{n}\sum_{i=1}^{n} X_i$ and $\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \left((\frac{1}{n})\sum_{i=1}^{n} X_i\right)^2$ are moment estimators of λ.

Example 5.2.5 Let $X_1, \ldots, X_n \sim \text{Poisson}(\lambda)$. Show that both $\frac{1}{n}\sum_{i=1}^{n} X_i$ and $\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \left((\frac{1}{n})\sum_{i=1}^{n} X_i\right)^2$ are moment estimators of λ.

Solution

- $E(X) = \lambda$

$$\hat{\lambda}_1 = \frac{1}{n}\sum_{i=1}^{n} X_i$$

- $Var(X) = \lambda$, $\lambda = E(X^2) - (EX)^2$, so that:

$$\hat{\lambda}_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)^2$$

- $\hat{\lambda}_1$ and $\hat{\lambda}_2$ are moment estimators of λ
- The moment estimators may not be unique
- We generally choose $\bar{X}$ as an estimator of λ for its simplicity

5.2.2 The method of maximum likelihood

Definition 5.2.1 (The likelihood function) let $f(x_1, \dots, x_n; \theta), \theta \in \Theta$ be the joint probability (or density) function of n random variables $X_1, \dots, X_n$ with sample values $x_1, \dots, x_n$. The likelihood function of the sample is given by:

$$L(\theta; x_1, \dots, x_n) = f(x_1, \dots, x_n; \theta) \triangleq L(\theta)$$

- For fixed sample values, L is a function of θ
- The likelihood of a set of parameter values θ, given $x_1, \dots, x_n$, is equal to the probability of those observed outcomes given the parameter values

5.2.2 The method of maximum likelihood

The likelihood function

- If $X_1, \ldots, X_n$ are discrete iid random variables with probability function $p(x;\theta)$, then the likelihood function is given by:

$$L(\theta) = P(X_1 = x_1, \ldots, X_n = x_n) = \prod_{i=1}^{n} P(X_i = x_i) = \prod_{i=1}^{n} p(x_i;\theta)$$

- In the continuous case, if the density is $f(x;\theta)$, then the likelihood function is:

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta)$$

Example 5.2.6 Let $X_1, \dots, X_n$ be iid $N(\mu, \sigma^2)$ random variables. Let $X_1, \dots, X_n$ be the sample values. Find the likelihood function.

Example 5.2.6 Let $X_1, \ldots, X_n$ be iid $N(\mu, \sigma^2)$ random variables. Let $X_1, \ldots, X_n$ be the sample values. Find the likelihood function.

Solution

The density function for the normal variable is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Hence, the likelihood function:

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

$$= \frac{1}{(2\pi)^{n/2}\sigma^n} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right)$$

5.2.2 The method of maximum likelihood

Definition 5.2.2 (Maximum likelihood estimators) Maximum likelihood estimators are those values of the parameters that maximize the likelihood function with respect to the parameter θ. That is,

$$L(\hat{\theta}; x_1, \dots, x_n) = \max_{\theta \in \Theta} L(\theta; x_1, \dots, x_n)$$

- Θ is the set of possible values of the parameter θ.
- The method of maximum likelihood extends to the case of several parameters:

$$\theta = (\theta_1, \dots, \theta_m)$$

5.2.2 The method of maximum likelihood

Procedure to find the maximum likelihood estimator

- Define the likelihood function, $L(\theta)$
- Often it is easier to take the natural logarithm $\ln$ of $L(\theta)$
- When applicable, differentiate $\ln L(\theta)$ with respect to θ, and then equate the derivative to zero
- Solve for the parameter θ, and we will obtain $\hat{\theta}$
- Check whether it is a maximizer or a global maximizer

5.2.2 The method of maximum likelihood

Example 5.2.7 Suppose $X_1, \ldots, X_n$ is a random sample from a geometric distribution with parameter p, $0 \le p \le 1$. Find the MLE $\hat{p}$.

Example 5.2.7 Suppose $X_1, \ldots, X_n$ is a random sample from a geometric distribution with parameter p, $0 \leq p \leq 1$. Find the MLE $\hat{p}$.

Solution

For the geometric distribution, the pmf is given by:

$$f(x, p) = p(1-p)^{x-1}, \quad 0 \leq p \leq 1, \quad x = 1, 2, 3, \ldots .$$

Hence, the likelihood function is:

$$L(p) = \prod_{i=1}^{n} \left[p(1-p)^{x_i - 1}\right] = p^n (1-p)^{-n + \sum_{i=1}^{n} x_i}.$$

Taking the natural logarithm of $L(p)$,

$$\ln L = n \ln p + \left(-n + \sum_{i=1}^{n} x_i\right) \ln(1-p)$$

Taking the derivative with respect to p, we have:

$$\frac{d \ln L}{dp} = \frac{n}{p} - \frac{-n + \sum_{i=1}^{n} x_i}{1-p}.$$

Equating $\frac{d \ln L(p)}{dp}$ to zero, we have:

$$\frac{n}{p} - \frac{\left(-n + \sum_{i=1}^{n} x_i\right)}{(1-p)} = 0.$$

Solving for p,

$$p = \frac{n}{\sum\limits_{i=1}^{n} x_i} = \frac{1}{\bar{x}}$$

Thus, we obtain a MLE of p as:

$$\hat{p} = \frac{n}{\sum\limits_{i=1}^{n} X_i} = \frac{1}{\bar{X}}.$$

Example 5.2.8

(a) Suppose $X_1, \ldots, X_n$ is a random sample from a Poisson distribution with parameter λ. Find MLE $\hat{\lambda}$.

(b) Traffic engineers use the Poisson distribution to model light traffic. This is based on the rationale that when the rate is approximately constant in light traffic, the distribution of counts of cars in a given time interval should be Poisson. The following data show the number of vehicles turning left in 15 randomly chosen 5-minute intervals at a specific intersection. Calculate the maximum likelihood estimate :

$$10, 17, 12, 6, 12, 11, 9, 6, 10, 8, 8, 16, 7, 10, 6.$$

Solution

(a) We have the probability mass function (pmf):

$$p(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \ldots, \ \lambda > 0.$$

Hence, the likelihood function is:

$$L(\lambda) = \prod_{i=1}^{n} \left(\frac{\lambda^{x_i} e^{-\lambda}}{x_i!} \right) = \frac{\lambda^{\sum\limits_{i=1}^{n} x_i} e^{-n\lambda}}{\prod\limits_{i=1}^{n} x_i!}$$

Then, taking the natural logarithm, we have:

$$\ln L(\lambda) = \sum_{i=1}^{n} x_i \ln(\lambda) - n\lambda - \sum_{i=1}^{n} \ln(x_i!)$$

and differentiating with respect to λ results in:

$$\frac{d \ln L(\lambda)}{d\lambda} = \frac{\sum_{i=1}^{n} x_i}{\lambda} - n \text{ and } \frac{d \ln L(\lambda)}{d\lambda} = 0, \text{ implies } \frac{\sum_{i=1}^{n} x_i}{\lambda} - n = 0.$$

That is,

$$\lambda = \frac{\sum_{i=1}^{n} x_i}{n} = \bar{x}.$$

Hence, MLE of λ is:

$$\hat{\lambda} = \bar{x}.$$

(b) From (a), we have the estimate as:

$$\hat{\lambda} = \bar{x} = 9.8;$$

or approximately 10 vehicles per 5 minutes turn left at this intersection.

5.2.2 The method of maximum likelihood

Example 5.2.9 Let $X_1, \ldots, X_n$ be a random sample from $U(0, \theta)$, $\theta > 0$. Find the MLE of θ.

5.2.2 The method of maximum likelihood

Example 5.2.9 Let $X_1, \ldots, X_n$ be a random sample from $U(0, \theta)$, $\theta > 0$. Find the MLE of θ. Solution

Note that the pdf of the uniform distribution is:

$$f(x) = \begin{cases} \frac{1}{\theta}, & 0 \leq x \leq \theta \\ 0, & \text{otherwise} \end{cases}$$

Hence, the likelihood function is given by:

$$L(\theta; x_1, x_2, \ldots, x_n) = \begin{cases} \frac{1}{\theta^n}, & 0 \leq x_1, x_2, \ldots, x_n \leq \theta \\ 0, & \text{otherwise} \end{cases}$$

5.2.2 The method of maximum likelihood

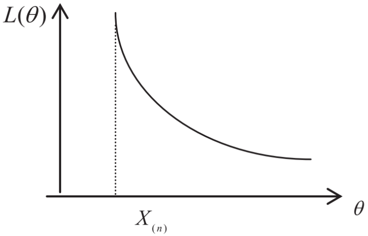


FIGURE 5.1 Likelihood function for uniform probability distribution.

- When $\theta \geq \max(X_i)$, the likelihood is $\frac{1}{\theta^n}$
- For $\theta < \max(X_i)$, the likelihood drops to 0
- We will not be able to find the derivative
- $\max(X_i)$ is the minimum value of θ that can be chosen that still satisfies the condition $0 \leq X_i \leq \theta$
- The MLE is the largest order statistic: $\hat{\theta} = \max(X_i) = X_{(n)}$

Example 5.2.10 Let $X_1, \ldots, X_n$ be $N\left(\mu, \sigma^2\right)$.
(a) If μ is unknown and $\sigma^2 = \sigma_0^2$ is known, find the MLE for μ.
(b) If $\mu = \mu_0$ is known and σ^2 is unknown, find the MLE for σ^2.
(c) If both μ and σ^2 are unknown, find the MLE for $\theta = (\mu, \sigma^2)$.

Example 5.2.10 Let $X_1, \ldots, X_n$ be $N(\mu, \sigma^2)$.
(a) If μ is unknown and $\sigma^2 = \sigma_0^2$ is known, find the MLE for μ.
(b) If $\mu = \mu_0$ is known and σ^2 is unknown, find the MLE for σ^2.
(c) If both μ and σ^2 are unknown, find the MLE for $\theta = (\mu, \sigma^2)$.

Solution

To avoid notational confusion when taking the derivative, let $\theta = \sigma^2$. Then, the likelihood function is:

$$L(\mu, \theta) = (2\pi\theta)^{-n/2} \exp\left(-\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{2\theta}\right)$$

or

$$\ln L(\mu, \theta) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\theta) - \frac{1}{2\theta}\sum_{i=1}^{n}(x_i - \mu)^2$$

(a) When $\theta = \theta_0 = \sigma_0^2$ is known, the problem reduces to estimating only one parameter, μ. Differentiating the log-likelihood function with respect to μ:

$$\frac{\partial}{\partial \mu}(\ln L(\mu, \theta_0)) = \frac{1}{\theta_0} \sum_{i=1}^{n} (x_i - \mu)$$

Setting the derivative equal to zero and solving for μ:

$$\sum_{i=1}^{n} (x_i - \mu) = 0$$

From this,

$$\sum_{i=1}^{n} x_i = n\mu \quad \text{or} \quad \mu = \bar{x}$$

Thus, we get $\hat{\mu} = \bar{X}$

(b) When $\mu = \mu_0$ is known, the problem reduces to estimating only one parameter, $\sigma^2 = \theta$.

5.2.2 The method of maximum likelihood

Differentiating the log-likelihood function with respect to θ:

$$\frac{d}{d\theta}\ln L(\mu,\theta) = \frac{-n}{2\theta} + \frac{1}{2\theta^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

Setting the derivative equal to zero and solving for θ, we get:

$$\hat{\theta} = \hat{\sigma}^2 = \frac{\sum_{i=1}^{n}(x_i-\mu_0)^2}{n}$$

(c) When both μ and θ are unknown, we need to differentiate with respect to both μ and θ individually:

$$\frac{d}{d\mu}\ln L(\mu,\theta) = \frac{1}{\theta}\sum_{i=1}^{n}(x_i-\mu) \text{ and } \frac{d}{d\theta}\ln L(\mu,\theta) = \frac{-n}{2\theta} + \frac{1}{2\theta^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

Setting the derivatives equal to zero and solving simultaneously, we obtain: $\hat{\mu} = \bar{X}; \quad \hat{\sigma}^2 = \hat{\theta} = \frac{1}{n}\sum_{i=1}^{n}\left(X_i-\bar{X}\right)^2 = {S'}^2$ Note that in (a) and (c), the estimates for μ are the same; however, in (b) and (c), the estimates for θ are different.

Example 5.2.11 Let $X_1, \ldots, X_n$ be a random sample from a population with a gamma distribution and parameters α and β. Find the MLEs for the unknown parameters α and β.

Example 5.2.11 Let $X_1, \dots, X_n$ be a random sample from a population with a gamma distribution and parameters α and β. Find the MLEs for the unknown parameters α and β.

Solution

The pdf for the gamma distribution is given by:

$$f(x) = \begin{cases} \frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\beta^{\alpha}} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases} \quad \alpha > 0, \beta > 0$$

The likelihood function is given by:

$$L(\alpha, \beta) = \frac{1}{(\Gamma(\alpha)\beta^{\alpha})^n} \prod_{i=1}^{n} x_i^{\alpha-1} e^{-\sum_{i=1}^{n} x_i/\beta}$$

Taking the logarithms gives:

$$\ln L = -n \ln \Gamma(\alpha) - n\alpha \ln \beta + (\alpha - 1) \sum_{i=1}^{n} \ln x_i - \sum_{i=1}^{n} \frac{x_i}{\beta}$$

Now, taking the partial derivatives with respect to α and β and setting both equal to zero, we have:

$$\frac{\partial}{\partial \alpha} \ln L = n \frac{\Gamma'(\alpha)}{\Gamma(\alpha)} - n \ln \beta + \sum_{i=1}^{n} \ln x_i = 0$$

$$\frac{\partial}{\partial \beta} \ln L = -n \frac{\alpha}{\beta} + \sum_{i=1}^{n} \frac{x_i}{\beta^2} = 0$$

Solving the second equation to get β in terms of α, we have: $\beta = \frac{\bar{x}}{\alpha}$

Substituting this β into the first equation, we need to solve:

$$-n\frac{\Gamma'(\alpha)}{\Gamma(\alpha)} - n\ln\frac{\bar{x}}{\alpha} + \sum_{i=1}^{n}\ln x_i = 0$$

for $\alpha > 0$. There is no closed-form solution for α and β. In this case, one can use numerical methods such as the Newton-Raphson method to solve for α, and then use this value to find β.

5.2.2.1 Some additional probability distributions

- Three-parameter gamma pdf
- Weibull pdf
- Rayleigh pdf
- Power exponential pdf
- Each of these pdfs will be applied to real data
  - cancer data, hurricane data, global warming data, and environmental (rainfall) data

5.2.2.1 Some additional probability distributions

Three-parameter gamma pdf

- The three-parameter gamma pdf is given by:

$$f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} (x-\gamma)^{\alpha-1} \exp\left(-\frac{(x-\gamma)}{\beta}\right)$$

where $x > \gamma$,$\beta > 0$, and $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$

- The corresponding cumulative distribution function (cdf) is given by:

$$F(x) = P(X \leq x) = \int_\gamma^x \frac{1}{\beta^\alpha \Gamma(\alpha)} (y-\gamma)^{\alpha-1} \exp\left(-\frac{(y-\gamma)}{\beta}\right) dy$$

$$= \Gamma_{\frac{x-\gamma}{\beta}}(\alpha)$$

- The expected value is given by:

$$E(X) = \int_0^\infty x f(x) dx = \gamma + \alpha\beta$$

5.2.2.1 Some additional probability distributions

Example 5.2.12 Given a random sample, $X_1, \dots, X_n$ from a three-parameter gamma distribution, obtain the MLEs of the parameters.

5.2.2.1 Some additional probability distributions

Example 5.2.12 Given a random sample, $X_1, \ldots, X_n$ from a three-parameter gamma distribution, obtain the MLEs of the parameters.

Solution

The likelihood function is given by:

$$\begin{aligned} L(\alpha, \beta, \gamma) &= \prod_{i=1}^{n} f(x_i) \\ &= \left(\frac{1}{\beta^{\alpha}\Gamma(\alpha)}\right)^{n} \prod_{i=1}^{n} (x_i - \gamma)^{\alpha-1} \prod_{i=1}^{n} \exp\left\{-\left(\frac{x_i - \gamma}{\beta}\right)\right\} \end{aligned}$$

and the log-likelihood function $\ell\left(\alpha, \beta, \gamma\right)$ of $L(\alpha, \beta, \gamma)$ is given by:

$$\ell(\alpha, \beta, \gamma) = -n\alpha \ln \beta - n \ln \Gamma(\alpha) + (\alpha - 1) \sum_{i=1}^{n} \ln(x_i - \gamma) - \sum_{i=1}^{n} \frac{x_i - \gamma}{\beta}$$

The MLEs can be obtained by setting $\frac{\partial \ell}{\partial \alpha} = 0, \frac{\partial \ell}{\partial \beta} = 0, \frac{\partial \ell}{\partial \gamma} = 0$

$$\frac{\partial \ell}{\partial \beta} = -\frac{n\alpha}{\beta} + \frac{\sum_{i=1}^{n}(x_i - \gamma)}{\beta^2} = 0,$$

which results in the MLE of β being:

$$\hat{\beta} = \frac{\sum_{i=1}^{n}(x_i - \hat{\gamma})}{n\hat{\alpha}}$$

$$\frac{\partial \ell}{\partial \alpha} = -n \ln \beta - n\frac{\Gamma'(\alpha)}{\Gamma(\alpha)} + \sum_{i=1}^{n} \ln(x_i - \gamma) = 0.$$

Substituting $\hat{\beta}$ in the equation for α, we have:

$$\ln(\hat{\alpha}) - \frac{\Gamma'(\alpha)}{\Gamma(\alpha)} = \ln\left[\frac{1}{n}\sum_{i=1}^{n}(x_1 - \hat{\gamma})\right] - \frac{1}{n}\sum_{i=1}^{n}\ln(x_i - \hat{\gamma}),$$

where $\frac{\Gamma'(\alpha)}{\Gamma(\alpha)}$ is called the digamma function, defined as the logarithmic derivative of the gamma function.Now,

$$\frac{\partial \ell}{\partial \gamma} = -(\alpha - 1) \sum_{i=1}^{n} \frac{1}{(x_i - \gamma)} + \sum_{i=1}^{n} \frac{1}{\beta} = 0,$$

Which reduces to:

$$\sum_{i=1}^{n} \frac{1}{x_i - \hat{\gamma}} = \frac{n}{\hat{\beta}(\hat{\alpha} - 1)}$$

Thus, we can proceed to numerically solve these equations to obtain (numerically) approximate MLE values for $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\gamma}$, so that we can apply the subject pdf to real data.

Weibull pdf

The Weibull probability distribution is very important in characterizing the behavior of health, engineering, and environmental data, among others.

- The Weibull pdf is given by:

$$f(x) = \frac{\alpha}{\beta}\left(\frac{x-\gamma}{\beta}\right)^{\alpha-1} \exp\left[-\left(\frac{x-\gamma}{\beta}\right)^{\alpha}\right]$$

  where $x > 0$, and the shape parameter $\alpha > 0$, the scale parameter $\beta > 0$ and the location parameter $\gamma < x$

- The cumulative probability distribution of the Weibull pdf is given by:

$$F(x) = 1 - \exp\left[-\left(\frac{x-\gamma}{\beta}\right)^{\alpha}\right]$$

5.2.2.1 Some additional probability distributions

Example 5.2.13 For a random sample $x_1, \ldots, x_n$ drawn from the three-parameter Weibull pdf, obtain MLEs for the parameters.

Example 5.2.13 For a random sample $x_1, \ldots, x_n$ drawn from the three-parameter Weibull pdf, obtain MLEs for the parameters.

Solution

The likelihood function, $L(\alpha, \beta, \gamma)$, is given by:

$$L(\alpha, \beta, \gamma) = \alpha^n \beta^{-n\alpha} \{\prod_{i=1}^{n} (x_i - \gamma)\}^{\alpha - 1} \exp\{-\beta^{-\alpha} \sum_{i=1}^{n} (x_i - \gamma)^\alpha\}$$

and the log-likelihood function $\ell\,(\alpha, \beta, \gamma)$ of $L(\alpha, \beta, \gamma)$ is given by:

$$\ell\,(\alpha, \beta, \gamma) = n \ln \alpha - n\alpha \ln \beta + (\alpha - 1) \sum_{i=1}^{n} \ln(x_i - \gamma) - \beta^{-\alpha} \sum_{i=1}^{n} (x_i - \gamma)^\alpha$$

Setting $\frac{\partial \ell}{\partial \alpha} = 0, \frac{\partial \ell}{\partial \beta} = 0$ and $\frac{\partial \ell}{\partial \gamma} = 0$, and taking the partial derivatives and substituting $\alpha = \hat{\alpha}$, $\beta = \hat{\beta}$, and $\gamma = \hat{\gamma}$, and simplifying the resulting expression, we have:

$$\hat{\alpha} + \sum_{i=1}^{n} \ln\left(x_i - \hat{\gamma}\right) = \frac{n \sum_{i=1}^{n} \left(x_i - \hat{\gamma}\right)^{\hat{\alpha}} \ln(x_i - \hat{\gamma})}{\sum_{i=1}^{n} (x_i - \hat{\gamma})^{\hat{\alpha}}},$$

$$\frac{n\hat{\alpha} \sum_{i=1}^{n} (x_i - \hat{\gamma})^{\hat{\alpha}-1}}{\sum_{i=1}^{n} (x_i - \hat{\gamma})^{\hat{\alpha}}} = (\hat{\alpha} - 1) \sum_{i=1}^{n} \frac{1}{x_i - \hat{\gamma}}$$

and

$$\hat{\beta} = \left\{ \frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{\gamma})^{\hat{\alpha}} \right\}^{\frac{1}{\hat{\alpha}}}$$

The above equation cannot be analytically solved without further restrictions, so we cannot obtain exact values for $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\gamma}$; however, there are software packages that we can use to obtain approximate estimates of the subject parameters.

5.2.2.1 Some additional probability distributions

Exponential power or error probability distribution

- It is usually applicable in characterizing continuous data that are very nonsymmetric with respect to their mean
- It is useful in analyzing environmental, engineering, and health data, among others.
- The power exponential or error pdf is given by:

$$f(x) = \lambda \left(e^{1-e^{\lambda x^k}}\right) e^{\lambda x^k} x^{k-1}, x > 0$$

where $\lambda > 0, k > 0$ are location and shape parameters, respectively

- The cumulative probability distribution function of the random variable X that follows the exponential power pdf is given by:

$$F(x) = 1 - e^{1-e^{\lambda x^k}}$$

Rayleigh distribution

- It characterizes the behavior of a continuous random variable that represents many real-world problems
- This pdf arises when there is a two-dimensional vector, for example, wind velocity data as measured by an anemometer and wind range that consists of speed value and direction, and both components are normally distributed, are not correlated, and have equal variance.
- The Rayleigh pdf of the random variable X is given by:

$$f(x;\sigma) = \frac{x}{\sigma^2} e^{\left(\frac{-x^2}{2\sigma^2}\right)}, x > 0$$

  where the scale parameter $\sigma > 0$
- The cdf is given by:

$$F(x) = P(X \leq x) = \int_0^x \frac{t}{\sigma^2} e^{\left(\frac{-t^2}{2\sigma^2}\right)} dt = 1 - e^{\left(\frac{-x^2}{2\sigma^2}\right)}$$

- The expected value and the variance are given by:

$$E(X) = \sigma\sqrt{\frac{\pi}{2}} = 1.25\sigma, Var(X) = \frac{4-\pi}{2}\sigma^2$$

- For a random sample $X_1, \ldots, X_n$ from the Rayleigh pdf, we can verify that the MLE of σ is given by

$$\hat{\sigma} = \frac{1}{2n}\sum_{i=1}^{n} X_i^2$$

5.2.2.1 Some additional probability distributions

Theorem 5.2.1 Let $h(\theta)$ be a one-to-one function of θ. If $\hat{\theta} = (\hat{\theta}_1, \ldots, \hat{\theta}_l)$ is the MLE of $\theta = (\theta_1, \ldots, \theta_l)$, then the MLE of a function

$$h(\theta) = (h_1(\theta), \ldots, h_k(\theta))$$

of these parameters is

$$h(\hat{\theta}) = (h_1(\hat{\theta}), \ldots, h_k(\hat{\theta}))$$

for $1 \leq k \leq l$.

5.3 Some desirable properties of point estimators

- It is possible to have several estimators for the same parameter.
- For a practitioner of statistics, an important question is going to be which of many available sample statistics, such as mean, median, smallest observation, or largest observation, should be chosen to represent all of the sample? Should we use the method of moments estimator, the MLE, or an estimator obtained through some other method such as the least squares (we will see this method in Chapter 7)?
- In this section, we will introduce some common ways to distinguish between them by looking at some desirable properties.

5.3.1 Unbiased estimators

Definition 5.3.1 A point estimator $\hat{\theta}$ is called an unbiased estimator of the parameter θ if $E(\hat{\theta}) = \theta$ for all possible values of θ. Otherwise $\hat{\theta}$ is said to be biased. Furthermore, the bias of $\hat{\theta}$ is given by:

$$B = E(\hat{\theta}) - \theta$$

Theorem 5.3.1 The mean of a random sample $\bar{X}$ is an unbiased estimator of the population mean μ

Theorem 5.3.2 If S^2 is the variance of a random sample from an infinite population with finite variance σ^2, then S^2 is an unbiased estimator for σ^2

5.3.1 Unbiased estimators

Example 5.3.1 Let $X_1, \ldots, X_n$ be a random sample from a Bernoulli population with parameter p. Show that the method of moments estimator is also an unbiased estimator.

Example 5.3.1 Let $X_1, \ldots, X_n$ be a random sample from a Bernoulli population with parameter p. Show that the method of moments estimator is also an unbiased estimator.

Solution

We can verify that the moment estimator of p is:

$$\hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{Y}{n}$$

Because for binomial random variables, $E(Y) = np$, it follows that:

$$E(\hat{p}) = E\left(\frac{Y}{n}\right) = \frac{1}{n}E(Y) = \frac{1}{n} \cdot np = p$$

Hence, $\hat{p} = \frac{Y}{n}$ is an unbiased estimator for p.

Example 5.3.2 Let $X_1, \dots, X_n$ be a random sample from a population with finite mean μ. Show that the sample mean $\bar{X}$ and $\frac{1}{3}\bar{X} + \frac{2}{3}X_1$ are both unbiased estimators of μ.

Example 5.3.2 Let $X_1, \ldots, X_n$ be a random sample from a population with finite mean μ. Show that the sample mean $\bar{X}$ and $\frac{1}{3}\bar{X} + \frac{2}{3}X_1$ are both unbiased estimators of μ. Solution

By Theorem 5.3.1, $\bar{X}$ is unbiased. Now:

$$E\left[\frac{1}{3}\bar{X} + \frac{2}{3}X_1\right] = \frac{1}{3}\mu + \frac{2}{3}\mu = \mu.$$

Hence $\frac{1}{3}\bar{X} + \frac{2}{3}X_1$ is also an unbiased estimator of μ.

5.3.1 Unbiased estimators

Example 5.3.3 Let $\hat{\theta}_1$ and $\hat{\theta}_2$ be two unbiased estimators of θ. Show that: $\hat{\theta}_3 = a\hat{\theta}_1 + (1-a)\hat{\theta}_2; \quad 0 \leq a \leq 1$ is an unbiased estimator of θ. Note that $\hat{\theta}_3$ is a convex combination of $\hat{\theta}_1$ and $\hat{\theta}_2$. In addition, assume that $\hat{\theta}_1$ and $\hat{\theta}_2$ are independent, and that $Var(\hat{\theta}_1) = \sigma_1^2$ and $Var(\hat{\theta}_2) = \sigma_2^2$. How should the constant a be chosen to minimize the variance of $\hat{\theta}_3$?

Example 5.3.3 Let $\hat{\theta}_1$ and $\hat{\theta}_2$ be two unbiased estimators of θ. Show that: $\hat{\theta}_3 = a\hat{\theta}_1 + (1-a)\hat{\theta}_2; \quad 0 \leq a \leq 1$ is an unbiased estimator of θ. Note that $\hat{\theta}_3$ is a convex combination of $\hat{\theta}_1$ and $\hat{\theta}_2$. In addition, assume that $\hat{\theta}_1$ and $\hat{\theta}_2$ are independent, and that $Var(\hat{\theta}_1) = \sigma_1^2$ and $Var(\hat{\theta}_2) = \sigma_2^2$. How should the constant a be chosen to minimize the variance of $\hat{\theta}_3$?

Solution

We are given that $E[\hat{\theta}_1] = \theta$ and $E[\hat{\theta}_2] = \theta$. Therefore,

$E[\hat{\theta}_3] = E\left[a\hat{\theta}_1 + (1-a)\hat{\theta}_2\right] = aE[\hat{\theta}_1] + (1-a)E[\hat{\theta}_2] = a\theta + (1-a)\theta = \theta$

Hence, $\hat{\theta}_3$ is unbiased. By independence,

$Var[\hat{\theta}_3] = Var\left[a\hat{\theta}_1 + (1-a)\hat{\theta}_2\right] = a^2 Var[\hat{\theta}_1] + (1-a)^2 Var[\hat{\theta}_2] = a^2\sigma_1^2 + (1-a)^2\sigma_2^2$

To find the minimum,

$$\frac{d}{da} Var[\hat{\theta}_3] = 2a\sigma_1^2 - 2(1-a)\sigma_2^2 = 0$$

5.3.1 Unbiased estimators

gives us: $a = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$

Because $\frac{d^2}{da^2} Var[\hat{\theta}_3] = 2\sigma_1^2 + 2\sigma_2^2 > 0$, $Var[\hat{\theta}_3]$ has a minimum at this value of $'a'$. Thus, if $\sigma_1^2 = \sigma_2^2$, then $a = \frac{1}{2}$.

Example 5.3.4 Let $X_1, \dots, X_n$ be a random sample from a population with pdf:

$$f(x) = \begin{cases} \frac{1}{\beta} e^{\frac{-x}{\beta}}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$$

Show that the method of moments estimator for the population parameter β is unbiased.

gives us: $a = \frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}$

Because $\frac{d^2}{da^2} Var[\hat{\theta}_3] = 2\sigma_1^2 + 2\sigma_2^2 > 0$, $Var[\hat{\theta}_3]$ has a minimum at this value of $'a'$. Thus, if $\sigma_1^2 = \sigma_2^2$, then $a = \frac{1}{2}$.

Example 5.3.4 Let $X_1, \dots, X_n$ be a random sample from a population with pdf:

$$f(x) = \begin{cases} \frac{1}{\beta} e^{\frac{-x}{\beta}}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$$

Show that the method of moments estimator for the population parameter β is unbiased.

Solution

From Section 5.2, we have seen that the method of moments estimator for β is the sample mean $\bar{X}$, and the population mean is β. Because $E(\bar{X}) = \mu = \beta$; the method of moments estimator for the population parameter β is unbiased.

Definition 5.3.2 The mean square error of the estimator $\hat{\theta}$, denoted by $MSE(\hat{\theta})$, is defined as

$$MSE(\hat{\theta}) = E\left(\hat{\theta} - \theta\right)^2$$

- The MSE is a measure that combines both bias and variance

$$\begin{aligned}
\mathrm{MSE}(\hat{\theta}) &= E(\hat{\theta} - \theta)^2 = E[(\hat{\theta} - E(\hat{\theta})) + (E(\hat{\theta}) - \theta)]^2 \\
&= E\left[(\hat{\theta} - E(\hat{\theta}))^2 + (E(\hat{\theta}) - \theta)^2 + 2(\hat{\theta} - E(\hat{\theta}))(E(\hat{\theta}) - \theta)\right] \\
&= E(\hat{\theta} - E(\hat{\theta}))^2 + E(E(\hat{\theta}) - \theta)^2 + 2E(\hat{\theta} - E(\hat{\theta}))(E(\hat{\theta}) - \theta) \\
&= \mathrm{Var}(\hat{\theta}) + [E(\hat{\theta}) - \theta]^2
\end{aligned}$$

  - $B = E(\hat{\theta}) - \theta$: the bias of the estimator

- The bias is zero for unbiased estimators, it is clear that

$$MSE(\hat{\theta}) = Var(\hat{\theta})$$

5.3.1 Unbiased estimators

Example 5.3.5 Let X_1, X_2, X_3 be a sample of size $n = 3$ from a distribution with an unknown mean μ, $-\infty < \mu < \infty$, where the variance σ^2 is a known positive number. Show that both $\hat{\theta}_1 = \bar{X}$ and $\hat{\theta}_2 = \frac{1}{8}(2X_1 + X_2 + 5X_3)$ are unbiased estimators for μ. Compare the variances of $\hat{\theta}_1$ and $\hat{\theta}_2$.

5.3.1 Unbiased estimators

Example 5.3.5 Let X_1, X_2, X_3 be a sample of size $n = 3$ from a distribution with an unknown mean μ, $-\infty < \mu < \infty$, where the variance σ^2 is a known positive number. Show that both $\hat{\theta}_1 = \bar{X}$ and $\hat{\theta}_2 = \frac{1}{8}(2X_1 + X_2 + 5X_3)$ are unbiased estimators for μ. Compare the variances of $\hat{\theta}_1$ and $\hat{\theta}_2$.

Solution

We have: $E(\hat{\theta}_1) = E(\bar{X}) = \frac{1}{3} \cdot 3\mu = \mu$

and $E(\hat{\theta}_2) = \frac{1}{8}\left[2EX_1 + EX_2 + 5EX_3\right] = \frac{1}{8}\left[2\mu + \mu + 5\mu\right] = \mu.$

Hence, both $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimator.

However, $Var(\hat{\theta}_1) = \frac{\sigma^2}{3}$,

whereas $Var(\hat{\theta}_2) = Var(\frac{2X_1+X_2+5X_3}{8}) = \frac{4}{64}\sigma^2 + \frac{1}{64}\sigma^2 + \frac{25}{64}\sigma^2 = \frac{30}{64}\sigma^2$

Because $Var(\hat{\theta}_1) < Var(\hat{\theta}_2)$, we see that $\bar{X}$ is a better unbiased estimator in the sense that the variance of $\bar{X}$ is smaller.

5.3.2 Sufficiency

Definition 5.3.4 Let $X_1, \ldots, X_n$ be a random sample from a probability distribution with unknown parameter θ. Then, the statistic $U = g(X_1, \ldots, X_n)$ is said to be sufficient for θ if the conditional pdf or pmf of $X_1, \ldots, X_n$ given $U = u$ does not depend on θ for any value of u.

- An estimator of θ that is a function of a sufficient statistic for θ is said to be a sufficient estimator of θ

Example 5.3.6 Let $X_1, \dots, X_n$ be iid Bernoulli random variables with parameter θ. Show that $U = \sum_{i=1}^{n} X_i$ is sufficient for θ.

Example 5.3.6 Let $X_1, \dots, X_n$ be iid Bernoulli random variables with parameter θ. Show that $U = \sum_{i=1}^{n} X_i$ is sufficient for θ.

Solution

The joint pmf of $X_1, \dots, X_n$ is:

$$f(x_1, \dots, x_n; \theta) = \theta^{\sum_{i=1}^{n} x_i}(1-\theta)^{n-\sum_{i=1}^{n} x_i}, \quad 0 \le \theta \le 1$$

Because $U = \sum_{i=1}^{n} X_i$ and $u = \sum_{i=1}^{n} x_i$, we have:

$$f(u_1, \dots, u_n; \theta) = \theta^{u}(1-\theta)^{n-u}, \quad 0 \le u \le n$$

Also, because $U \sim B(n, \theta)$, we have:

$$f(u; \theta) = \binom{n}{u}\theta^{u}(1-\theta)^{n-u}$$

5.3.2 Sufficiency

Also,

$$f(x_1,\dots,x_n|U=u)=\frac{f(x_1,\dots,x_n;u)}{f_U(u)}=\begin{cases}\frac{f(x_1,\dots,x_n)}{f_U(u)}, & u=\sum x_i\\ 0 & \text{otherwise}\end{cases}$$

Therefore,

$$f(x_1,\dots,x_n|U=u)=\begin{cases}\frac{\theta^u(1-\theta)^{n-u}}{\binom{n}{u}\theta^u(1-\theta)^{n-u}}=\frac{1}{\binom{n}{u}} & \text{if } u=\sum x_i\\ 0, & \text{otherwise.}\end{cases}$$

which is independent of θ. Therefore, U is sufficient for θ.

- Example 5.3.7
  Let, $X_1, \ldots, X_n$ be a random sample from $U(0, \theta)$. That is,

$$f(x) = \begin{cases} \frac{1}{\theta}, & \text{if } 0 < x < \theta \\ 0, & \text{otherwise.} \end{cases}$$

  Show that $U = \max_{1 \leq i \leq n} X_i$ is sufficient for θ.

- Solution
  The joint density or the likelihood function is given by:

$$f(x_1, \ldots, x_n; \theta) = \begin{cases} \frac{1}{\theta^n}, & \text{if } 0 < x_1, \ldots, x_n < \theta \\ 0, & \text{otherwise.} \end{cases}$$

  The joint pdf $f(x_1, \ldots, x_n; \theta)$ can be equivalently written as:

$$f(x_1, \ldots, x_n; \theta) = \begin{cases} \frac{1}{\theta^n}, & \text{if } x_{min} > 0, x_{max} < \theta \\ 0, & \text{otherwise.} \end{cases}$$

Now, we can compute the pdf of U:

$$F(u) = P(U \leq u) = P(X_1, \ldots, X_n \leq u)$$

$$= \Pi_{i=1}^{n} P(X_i \leq u) \text{ (because of independence)}$$

$$= \Pi_{i=1}^{n} \left(\int_0^u \frac{1}{\theta} dx \right) = \frac{u^n}{\theta^n}, 0 < u < \theta$$

The pdf of U may now be obtained as:

$$f(u) = \frac{d}{du} F(u) = \frac{n u^{n-1}}{\theta^n}, 0 < n < \theta$$

Moreover,

$$f(x_1, \ldots, x_n | u) = \begin{cases} \frac{f(x_1, \ldots, x_n, u)}{f_U(u)} = \frac{f(x_1, \ldots, x_n)}{f_U(u)}, & \text{if } x_{max} \text{ and } x_{min} > 0 \\ 0, & \text{otherwise} \end{cases}$$

Using the expressions for $f(x_1, \ldots, x_n)$ and $F_U(u)$ we obtain:

$$f(x_1, \ldots, x_n | u = u) = \begin{cases} \frac{1/\theta^n}{nu^{n-1}/\theta^n} = \frac{1}{nu^{n-1}}, & \text{if } x_{max} \text{ and } x_{min} > 0 \\ 0, & \text{otherwise} \end{cases}$$

$f(X_1, \ldots, X_n | U)$ is a function of u and x_{min}, which is independent of θ. Hence, $U = max_{1 \leq i \leq n} X_i$ is sufficient for θ.

5.3.2 Sufficiency

- Neyman-Fisher factorization criteria
  Let U be a statistic based on the random sample $X_1, \dots, X_n$. Then, U is a sufficient statistic for θ if and only if the joint pdf (or pf) $f(x_1, \dots, x_n; \theta)$ (which depends on the parameter θ) can be factored into two nonnegative functions:

$$f(x_1, \dots, x_n; \theta) = g(u, \theta) h(x_1, \dots, x_n), for\ all\ x_1, \dots, x_n$$

  where $g(u, \theta)$ is a function only of u and θ and $h(x_1, \dots, x_n)$ is a function of only $x_1, \dots, x_n$ and not of θ

- Procedure to verify sufficiency
  - Obtain the joint pdf or pf $f_\theta(x_1, \dots, x_n)$
  - If necessary, rewrite the joint pdf or pf in terms of the given statistic and parameter so that one can use the factorization theorem
  - Define the functions g and h in such a way that g is a function of the statistic and parameter only and h is a function of the observations only
  - If step 3 is possible, then the statistic is sufficient. Otherwise, it is not sufficient
- Example 5.3.8 and Example 5.3.9

- Example 5.3.8
  Let $X_1, \ldots, X_n$ denote a random sample from a geometric population with parameter p. Show that $\bar{X}$ is sufficient for p.

- Solution
  For the geometric distribution, the pdf is given by:

$$f(x,p) = \begin{cases} p(1-p)^{x-1}, & x \geq 1 \\ 0, & \text{otherwise} \end{cases}$$

Hence, the joint pf is:

$$\begin{aligned} f(x_1, \ldots, x_n) &= p^n (1-p)^{-n+\sum_{i=1}^n x_i} \\ &= \begin{cases} p^n (1-p)^{n\bar{x}-n}, & \text{if } x_1, \ldots, x_n \geq 1 \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

Take,

$$g(\bar{x}, p) = p^n (1-p)^{n\bar{x}-n}$$

and

$$h(x_1, \dots, x_n) = \begin{cases} 1, & \text{if } x_i \geq 1 \\ 0, & \text{otherwise} \end{cases}$$

Thus, $\bar{X}$ is sufficient for p.

- Example 5.3.9

  Let $X_1, \dots, X_n$ denote a random sample from a $U(0, \theta)$ with pdf:

$$f_\theta(x) = \begin{cases} \frac{1}{\theta}, & 0 < x < \theta, \theta > 0 \\ 0, & \text{otherwise} \end{cases}$$

  Show that $X_{(n)} = max_{1 \le i \le n} X_i$ is sufficient for θ, using the factorization theorem.

- Solution

  The likelihood function of the sample is:

$$f_\theta(x_1, \dots, x_n) = \begin{cases} \frac{1}{\theta^n}, & \text{if } 0 < x_1, \dots, x_n < \theta \\ 0, & \text{otherwise} \end{cases}$$

5.3.2 Sufficiency

We can now write $f_\theta(x_1, \ldots, x_n)$ as:

$$f_\theta(x_1, \ldots, x_n) = h(x_1, \ldots, x_n) g(\theta, x_{(n)}), \text{ for all } x_1, \ldots, x_n$$

where

$$h(x_1, \ldots, x_n) = \begin{cases} 1, & \text{if } x_1, \ldots x_n > 0 \\ 0, & \text{otherwise} \end{cases}$$

and

$$g(\theta; x_{(n)}) = \begin{cases} \frac{1}{\theta^n}, & \text{if } 0 < x_{(n)} < \theta \\ 0, & \text{otherwise} \end{cases}$$

From the factorization theorem, we now conclude that $X_{(n)}$ is sufficient for θ.

5.3.2 Sufficiency

Definition 5.3.5

Two statistics U_1 and U_2 are said to be jointly sufficient for the parameters θ_1 and θ_2 if the conditional distribution of $X_1, \ldots, X_n$ given U_1 and U_2 does not depend on θ_1 or θ_2.
In general, the statistic $U = (U_1, \ldots, U_n)$ is jointly sufficient for $\theta = (\theta_1, \ldots, \theta_n)$ if the conditional distribution of $X_1, \ldots, X_n$ given U is free of θ.

5.3.2 Sufficiency

Theorem 5.3.4 The factorization criteria for joint sufficiency

Theorem 5.3.4

The two statistics U_1 and U_2 are jointly sufficient for θ_1 and θ_2 if and only if the likelihood function can be factored into two nonnegative functions,

$$f(x_1, \dots, x_n; \theta_1, \theta_2) = g(u_1, u_2; \theta_1, \theta_2)h(x_1, \dots, x_n)$$

where $g(u_1, u_2; \theta_1, \theta_2)$ is only a function of $u_1, u_2; \theta_1$ and θ_2, and $h(x_1, \dots, x_n)$ is free of θ_1 and θ_2

Example 5.3.10 and Example 5.3.11

- Example 5.3.10

  Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$.
  (a) If μ is unknown and $\sigma^2 = \sigma_0^2$ is known, show that $\bar{X}$ is a sufficient statistic for μ.
  (b) If $\mu = \mu_0$ is known and σ^2 is unknown, show that $\sum_{i=1}^n (X_i - \mu_0)^2$ is sufficient for σ^2 .
  (c) If μ and σ^2 are both unknown, show that $\sum_{i=1}^n X_i$ and $\sum_{i=1}^n X_i^2$ are jointly sufficient for μ and σ^2 .

- Solution

  The likelihood function of the sample is:

$$L = \frac{1}{(2\pi)^{n/2}\sigma^n} \exp\left[-\frac{\sum_{i=1}^n (X_i - \mu)^2}{2\sigma^2}\right]$$

$$= \frac{1}{(2\pi)^{n/2}\sigma^n} \exp\left[-\frac{1}{2\sigma^2}\left(\sum_{i=1}^n x_i^2 - 2\mu\sum_{i=1}^n x_i + n\mu^2\right)\right]$$

$$= (2\pi)^{-n/2}\sigma^{-n}\exp\left(-\frac{\sum_{i=1}^{n}x_i^2}{2\sigma^2}\right)\exp\left(\frac{2\mu n\bar{x}}{2\sigma^2}\right)\exp\left(-\frac{n\mu^2}{2\sigma}\right)$$

(a) When $\sigma^2 = \sigma_0^2$ is known, use the factorization criteria, with:

$$g(\bar{x},\mu) = \exp\left(\frac{2n\mu\bar{x} - n\mu^2}{2\sigma_0^2}\right)$$

and

$$h(x_1,\ldots,x_n) = (2\pi)^{-n/2}\sigma^{-n}\exp\left(-\frac{\sum_{i=1}^{n}x_i^2}{2\sigma^2}\right)$$

Therefore, $\bar{X}$ is sufficient for μ.
(b) When $\mu = \mu_0$ is known, let

$$g\left(\sum_{i=1}^{n}(x_i-\mu)^2, \sigma^2\right) = \sigma^{-n}\exp\left[-\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}\right]$$

and

$$h(x_1, \ldots, x_n) = \frac{1}{(2\pi)^{n/2}}$$

Thus, $\sum_{i=1}^n (X_i - \mu)^2$ is sufficient for σ^2.
(c) When both μ and σ^2 are unknown, use:

$$g\left(\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2, \mu, \sigma^2\right) = \sigma^{-n} \exp\left(-\frac{\sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2}{2\sigma^2}\right)$$

and

$$h(x_1, \ldots, x_n) = \frac{1}{(2\pi)^{n/2}}$$

Hence, $\sum_{i=1}^n X_i$ and $\sum_{i=1}^n X_i^2$ are jointly sufficient for μ and σ^2

- Example 5.3.11
  Suppose that we have a random sample $X_1, \dots, X_n$ from a discrete distribution given by:
  $$f_\theta(x) = C(\theta)2^{-x/\theta}, x = \theta, \theta+1, \theta+2, \dots; \theta > 0.$$
  where $C(\theta) > 0$ is a normalizing constant. Using the factorization theorem, find a sufficient statistic for θ.

- Solution
  The joint density function $f(x_1, \dots, x_n; \theta)$ of the sample $X_1, \dots, X_n$ is:
  $$f(x_i, \dots, x_n; \theta) = \begin{cases} C(\theta)2^{-\sum\limits_{i=1}^{n}(x_i/\theta)}, & x_1, x_2, \dots, x_n \text{ are integers } \geq \theta \\ 0, & \text{otherwise} \end{cases}$$
  The function $f(x_1, \dots, x_n; \theta)$ can be written as:
  $$f(x_1, \dots, x_n; \theta) = h(x_1, \dots, x_n)C(\theta)2^{-\sum_{i=1}^{n}(x_i/\theta)}g_1(\theta, x_{(1)}),$$

where $x_{(1)} = \min(x_1, \dots, x_n)$ and

$$h(x_1, x_2, \dots, x_n) = \begin{cases} 1, & \text{if } x_j - x_{(1)} \geq 0 \text{ is an integer for } j = 1, 2, \dots, n \\ 0, & \text{otherwise.} \end{cases}$$

and

$$g_1(\theta, x_{(1)}) = \begin{cases} 1, & \text{if } x_{(1)} \geq \theta \\ 0, & \text{otherwise.} \end{cases}$$

Thus,

$$f(x_1, \dots, x_n; \theta) = h(x_1, \dots, x_n) g\left(\theta, \sum x_i, x_{(1)}\right)$$

where $g(\theta, \sum x_i, x_{(1)}) = C(\theta)2 - \sum_{i=1}^{n}(x_i/\theta) g_1(\theta, x_{(1)})$. Using the factorization theorem, we conclude that $(\sum x_i, x_{(1)})$ is jointly for θ. This result shows that even for a single parameter, we may need more than one statistic for sufficiency

Theorem 5.3.5

If U is a sufficient statistic for θ, the MLE of θ, if unique, is function of U.

Example 5.3.12 Write the following in exponential form:

(a) $\frac{e^{-\lambda}\lambda^x}{x!}$
(b) $p^x(1-p)^{1-x}$
(c) $\frac{1}{\sqrt{2\pi}}e^{-(x-\mu)^2/2}$

Solution

(a) We have:

$$\frac{e^{-\lambda}\lambda^x}{x!} = \exp[x\ln\lambda - \ln x! - \lambda].$$

Here, $k(x) = x, c(\lambda) = \ln\lambda, S(x) = -\ln(x!)$ and $d(\lambda) = -\lambda$.

(b) Similarly,

$$p^x(1-p)^{1-x} = \exp\left[x\ln(\frac{p}{1-p}) + \ln(1-p)\right], x = 0 \text{ or } 1.$$

(c) This is the standard normal density:

$$\frac{1}{\sqrt{2\pi}}e^{-(x-\mu)^2/2} = \exp\left[x\mu - \frac{x^2}{2} - \frac{\mu^2}{2} - \frac{1}{2}\ln(2\pi)\right], -\infty < x < \infty.$$

5.3.2 Sufficiency

Theorem 5.3.6

Let $X_1, \ldots, X_n$ be a random sample from a population with pdf or pmf of the exponential form:

$$f(x;\theta) = \begin{cases} \exp\left[k(x)c(\theta) + S(x) + d(\theta)\right], \text{ if } x \in B \\ 0, \ \ x \notin B \end{cases}$$

where B does not depend on the parameter θ. The statistic $\sum_{i=1}^{n} k(X_i)$ is sufficient for θ.

5.3.2 Sufficiency

Theorem 5.3.7 Rao-Blackwell theorem

Let $X_1, \dots, X_n$ be a random sample with joint pf or pdf $f(x_1, \dots, x_n; \theta)$ and let $U = (U_1, \dots, U_n)$ be jointly sufficient for $\theta = (\theta_1, \dots, \theta_n)$. If T is any unbiased estimator of $k(\theta)$, and if $T^* = E(T|U)$, then:

(a) T^* is an unbiased estimator of $k(\theta)$

(b) T^* is a function of U and does not depend on θ

(c) $Var(T^*) \le Var(T)$ for every θ, and $Var(T^*) < Var(T)$ for some θ unless $T^* = T$ with probability 1

Contents

5.4 The pivotal method for finding the confidence interval

The interval estimation will convey any measure of reliability

- For an interval estimator of a single parameter θ, we will use the random sample to find two quantities L and U such that $L < \theta < U$ with some probability
- L and U depend on the sample values, so they will be random.
- This interval (L, U) should have two properties:
  - $P(L < \theta < U)$ is high, that is, the true parameter θ is in (L, U) with high probability
  - The length of the interval (L, U) should be relatively narrow on average.

5.4 The pivotal method for finding the confidence interval

- Interval estimation is to find the estimating interval (L, U)

$$P(L \leq \theta \leq U) = 1 - \alpha$$

- Interval estimators are called confidence intervals
- The limits are called U and L, the upper and lower confidence limits, respectively
- The probability that a CI will contain the true parameter θ, $1 - \alpha$, is called the confidence coefficient
- We are $(1 - \alpha)100\%$ confident that the true parameter θ is located in the interval (L, U).

5.4 The pivotal method for finding the confidence interval

Example 5.4.1 As part of a promotion, the management of a large health club wants to estimate average weight loss for its members within the first 3 months after joining the club. They took a random sample of 45 members of this health club and found that they lost an average of 13.8 lb within the first 3 months of membership with a sample standard deviation of 4.2 lb. Find a 95% CI for the true mean. What if a random sample of 200 members of this health club also resulted in the same sample mean and sample standard deviation?

Example 5.4.1 As part of a promotion, the management of a large health club wants to estimate average weight loss for its members within the first 3 months after joining the club. They took a random sample of 45 members of this health club and found that they lost an average of 13.8 lb within the first 3 months of membership with a sample standard deviation of 4.2 lb. Find a 95% CI for the true mean. What if a random sample of 200 members of this health club also resulted in the same sample mean and sample standard deviation?

Solution

- By CLT we can get $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$
- $P\left(z_{-\alpha/2} \leq \frac{\bar{X}-\mu}{0.626} \leq z_{\alpha/2}\right) = 1-\alpha$
- $P\left(\bar{X} - \sigma/\sqrt{n} \times z_{\alpha/2} \leq \mu \leq \bar{X} + \sigma/\sqrt{n} \times z_{\alpha/2}\right) = 1-\alpha$

5.4 The pivotal method for finding the confidence interval

Let $1-\alpha = 95\%$

- For $n = 45$, the CI is $13.8 \pm (1.96)(0.626) = (12.57, 15.03)$, with 95% confidence, one can expect the true mean to lie in this interval
- For $n = 200$, the standard error is $(4.2/\sqrt{200}) \approx 2.97$. Thus a 95% CI is $13.8 \pm (1.96)(0.297)$ resulting in the interval $(13.22, 14.38)$
- The more sample values (that is, the more information) we have, the tighter (smaller width) the interval.

The previous example was built on our knowledge of the sampling distribution of the sample mean. More generally, our success in building CIs for an estimate of a parameter depends on identifying a quantity known as the pivot.

5.4 The pivotal method for finding the confidence interval

Pivotal method: a general method of constructing a CI and relies on our knowledge of sampling distributions. We should find a pivotal quantity with the following two characteristics:

- It is a function of the random sample (a statistic or an estimator $\hat{\theta}$) and the unknown parameter θ, where θ is the only unknown quantity
- It has a probability distribution that does not depend on the parameter θ

5.4 The pivotal method for finding the confidence interval

Procedure to find a confidence interval for θ using the pivot

- Find an estimator $\hat{\theta}$ of θ: usually the MLE of θ works
- Find a function of θ and $\hat{\theta}$, $p(\theta, \hat{\theta})$(pivot), such that the probability distribution of $p(\cdot, \cdot)$ does not depend on θ
- Find a and b such that $P\Big(a \le p\Big(\theta, \hat{\theta}\Big) \le b\Big) = 1 - \alpha$.
- Choose a and b such that

$$P\left(p\left(\theta, \hat{\theta}\right) \le a\right) = \alpha/2 \quad \text{and} \quad P\left(p\left(\theta, \hat{\theta}\right) \ge b\right) = \alpha/2$$

- Transform the pivot CI to a CI for the parameter θ:

$$P(L \le \theta \le U) = 1 - \alpha$$

Example 5.4.2 Suppose we have a random sample $X_1, \ldots, X_n$ from $N(\mu, 1)$. Construct a 95% CI for μ

- $1 - \alpha = 0.95$
- The MLE of $\mu = \bar{X} \sim N(\mu, 1/n)$. It cannot be a pivot
- Z can be a pivot $p(\hat{\theta}, \theta)$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - \mu}{1/\sqrt{n}} \sim N(0,1)$$

- Now to find a and b such that

$$P\left(a \leq Z \leq b\right) = p\left(a \leq p(\hat{\theta}, \theta) \leq b\right) = 0.95$$

5.4 The pivotal method for finding the confidence interval

One such choice is to find the value of a such that $P(-a \leq Z \leq a) = 0.95$
From the normal table,

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 0.95,$$

This implies $z_{\alpha/2} = 1.96$. Hence, $P(-1.96 \leq Z \leq 1.96) = 0.95$ or, using the definition of Z and solving for μ, we obtain:

$$P\left(\bar{X} - \frac{1.96}{\sqrt{n}} \leq \mu \leq \bar{X} + \frac{1.96}{\sqrt{n}}\right) = 0.95.$$

- A 95% CI for μ is $(\bar{X} - (1.96/\sqrt{n}), \bar{X} + (1.96/\sqrt{n}))$
- The lower confidence limit L is $\bar{X} - (1.96/\sqrt{n})$
- The upper confidence limit U is $\bar{X} + (1.96/\sqrt{n})$

Example 5.4.3 Suppose the random sample $X_1, \dots, X_n$ has $U(0, \theta)$ distribution. Construct a 90% CI for θ and interpret. Identify the upper and lower confidence limits.

Example 5.4.3 Suppose the random sample $X_1, \dots, X_n$ has $U(0, \theta)$ distribution. Construct a 90% CI for θ and interpret. Identify the upper and lower confidence limits.

Solution

From Example 5.3.4, we know that:

$$U = \max_{1 \le i \le n} X_i$$

is the MLE of θ. *The random variable* U *has the pdf:*

$$f_U(u) = nu^{n-1}/\theta^n, \quad 0 \le u \le \theta.$$

This is not independent of the parameter θ. *Let* $Y = U/\theta$, *then (using the Jacobians described in Chapter 3) the pdf of* Y *is given by:*

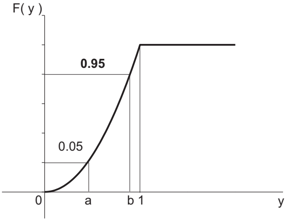
$$f_Y(y) = ny^{n-1}, \quad 0 \le y \le 1.$$

Hence, Y *satisfies the two characteristics of the pivotal quantity. Thus,* $Y = U/\theta$ *is a pivot. Now, we have to find* a *and* b *such that:*

$$p\left(a \le \frac{U}{\theta} \le b\right) = 0.90.$$

5.4 The pivotal method for finding the confidence interval

To find a *and* b *we use the cdf of* Y, $F_Y(y) = y^n$, $0 \le y \le 1$, *as follows:*



$$F_Y(a) = 0.05 \quad \text{and} \quad F_Y(b) = 0.95,$$

which implies that:

$$a^n = 0.05 \quad \text{and} \quad b^n = 0.95$$

resulting in:

$$a = \sqrt[n]{0.05} \quad \text{and} \quad b = \sqrt[n]{0.95}.$$

Write:

$$P\left(\sqrt[n]{0.05} < \frac{U}{\theta} < \sqrt[n]{0.95}\right) = 0.90.$$

Solving, the 90% CI for θ *is:*

$$\left(\frac{U}{\sqrt[n]{0.95}}, \frac{U}{\sqrt[n]{0.05}}\right)$$

or

$$P\left(\frac{U}{\sqrt[n]{0.95}} \le \theta \le \frac{U}{\sqrt[n]{0.05}}\right) = 0.90.$$

Thus, the lower confidence limit is $\mathrm{U}/\sqrt[n]{0.95}$ *and the upper confidence limit is* $\mathrm{U}/\sqrt[n]{0.05}$, *and the 90% CI is* $\left(\mathrm{U}/\sqrt[n]{0.95},\ \mathrm{U}/\sqrt[n]{0.05}\right)$

Contents

Find CIs for the one-sample case:

- Large sample situations
- Small-sample situations

5.5.1 Large-sample confidence intervals

Procedure to calculate large-sample confidence interval for θ

- Find an estimator (such as the MLE) of θ, say $\hat{\theta}$
- Obtain the standard error, $\sigma_{\hat{\theta}}$ of $\hat{\theta}$
- Find the z-transform $z = \dfrac{\left(\hat{\theta} - \theta\right)}{\sigma_{\hat{\theta}}}$. Then z has an approximately standard normal distribution
- Using the normal table, find two tail values $-z_{\frac{\alpha}{2}}$ and $z_{\frac{\alpha}{2}}$
- An approximate $(1-\alpha)100\%$ CI for θ is $\left(\hat{\theta} - z_{\frac{\alpha}{2}}\sigma_{\hat{\theta}}, \hat{\theta} + z_{\frac{\alpha}{2}}\sigma_{\hat{\theta}}\right)$, that is

$$P\left(\hat{\theta} - z_{\frac{\alpha}{2}}\sigma_{\hat{\theta}} \leq \theta \leq \hat{\theta} + z_{\frac{\alpha}{2}}\sigma_{\hat{\theta}}\right) = 1 - \alpha$$

5.5.1 Large-sample confidence intervals

Example 5.5.1 Let $\hat{\theta}$ be a statistic that is normally distributed with mean θ and standard deviation σ_{θ}, where σ_{θ} is assumed to be known. Find a CI for θ that possesses a confidence coefficient equal to $1-\alpha$.

5.5.1 Large-sample confidence intervals

Example 5.5.1 Let $\hat{\theta}$ be a statistic that is normally distributed with mean θ and standard deviation σ_θ, where σ_θ is assumed to be known. Find a CI for θ that possesses a confidence coefficient equal to $1-\alpha$.

Solution

The z-transform of $\hat{\theta}$ is, $Z = \frac{\hat{\theta}-\theta}{\sigma_\theta}$ and has a standard normal distribution. Select two tail values $-z_{\alpha/2}$ and $z_{\alpha/2}$ such that,

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1-\alpha$$

Because of symmetry, this is the shortest interval that contains the area $1-\alpha$. Then,

$$P(\hat{\theta} - z_{\alpha/2}\sigma_\theta \leq \theta \leq \hat{\theta} + z_{\alpha/2}\sigma_\theta) = 1-\alpha$$

Therefore, the confidence limit of θ are $\hat{\theta} - z_{\alpha/2}\sigma_\theta$ and $\hat{\theta} + z_{\alpha/2}\sigma_\theta$. Hence, the $(1-\alpha)100\%$ CI for θ is given by $\hat{\theta} \pm z_{\alpha/2}\sigma_\theta$.

5.5.1 Large-sample confidence intervals

Large-sample confidence interval

- For a large sample of size n, let $\hat{\theta} = \bar{X}$ be the sample mean. Then the large-sample $(1-\alpha)100\%$ CI for the population mean μ is:

$$\bar{X} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \simeq \bar{X} \pm z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}}$$

where S is a point estimate of σ. That is,

$$P\left(\bar{X} - z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

Example 5.5.2 Two statistics professors want to estimate average scores for an elementary statistics course that has two sections. Each professor teaches one section and each section has a large number of students. A random sample of 50 scores from each section produced the following results:

(a) Section I: $\bar{x}_1 = 77.01, s_1 = 10.32$

(b) Section II: $\bar{x}_2 = 72.22, s_2 = 11.02$

Calculate 95% CIs for each of these two samples.

Example 5.5.2 Two statistics professors want to estimate average scores for an elementary statistics course that has two sections. Each professor teaches one section and each section has a large number of students. A random sample of 50 scores from each section produced the following results:

(a) Section I: $\bar{x}_1 = 77.01, s_1 = 10.32$

(b) Section II: $\bar{x}_2 = 72.22, s_2 = 11.02$

Calculate 95% CIs for each of these two samples.

Solution

Because $n = 50$ is large, we could use normal approximation, For $\alpha = 0.05$, from the normal table: $z_{\alpha/2} = z_{0.025} = 1.96$. The CIs are as follows:

(a) We have:

$$\bar{x}_1 \pm z_{\alpha/2}\frac{s_1}{\sqrt{n}} = 77.01 \pm 1.96(\frac{10.32}{\sqrt{50}})$$

which gives a 95% CI$(74.149, 79.871)$.

(b) We can compute:

$$\bar{x}_2 \pm z_{\alpha/2}\frac{s_2}{\sqrt{n}} = 72.22 \pm 1.96(\frac{11.02}{\sqrt{50}})$$

which gives the interval $(69.165, 75.275)$.

5.5.3 Small-sample confidence intervals for μ

Theorem 5.5.1 If $\bar{X}$ and S are the sample mean and the sample standard deviation of a random sample of size n from a normal population, then:

$$\bar{X} - t_{\frac{\alpha}{2},n-1}\frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\frac{\alpha}{2},n-1}\frac{S}{\sqrt{n}}$$

is $(1-\alpha)100\%$ CI for the population mean μ.

5.5.3 Small-sample confidence intervals for μ

Procedure to find small-sample confidence interval for μ

- Calculate the values of $\bar{X}$ and S from the sample $X_1, \ldots, X_n$
- Using the t table, select two tail values, $-t_{\frac{\alpha}{2}}$ and $t_{\frac{\alpha}{2}}$
- The $(1-\alpha)100\%$ CI for μ is:

$$\bar{X} - t_{\frac{\alpha}{2}, n-1}\frac{S}{\sqrt{n}} \le \mu \le \bar{X} + t_{\frac{\alpha}{2}, n-1}\frac{S}{\sqrt{n}}$$

  that is,

$$P\left(\bar{X} - t_{\frac{\alpha}{2}, n-1}\frac{S}{\sqrt{n}} \le \mu \le \bar{X} + t_{\frac{\alpha}{2}, n-1}\frac{S}{\sqrt{n}}\right) = 1-\alpha$$

- Conclusion: We are $(1-\alpha)100\%$ confident that the true parameter μ lies in the interval

$$\bar{X} - t_{\frac{\alpha}{2}, n-1}\frac{S}{\sqrt{n}} \le \mu \le \bar{X} + t_{\frac{\alpha}{2}, n-1}\frac{S}{\sqrt{n}}$$

5.5.3 Small-sample confidence intervals for μ

Example 5.5.7 The following is a random data set from a normal population:

7.2 5.7 4.9 6.2 8.5 2.8

Construct a 95% CI for the population mean μ. Interpret.

5.5.3 Small-sample confidence intervals for μ

Example 5.5.7 The following is a random data set from a normal population:

7.2 5.7 4.9 6.2 8.5 2.8

Construct a 95% CI for the population mean μ. Interpret.

Solution

The first step is to calculate mean and standard deviation of the sample. We compute as the mean $\bar{X} = 5.883$ and as standard deviation,$s = 1.959$. For 5 degrees of freedom, and for $\alpha = 0.05$, from the t table, $t_{0.025} = 2.571$. Hence, a 95% CI for μ is:

$$\left(\bar{x} - t_{\alpha/2,n-1}\frac{2}{\sqrt{n}}, \bar{x} + t_{\alpha/2,n-1}\frac{2}{\sqrt{n}}\right)$$

$$= \left(5.883 - 2.571(\frac{1.959}{\sqrt{6}}), 5.5883 + 2.571(\frac{1.959}{\sqrt{6}})\right) = (3.827, 7.939).$$

This can be interpreted as that we are 95% confident that the true mean μ will be between 3.827 and 7.939.

5.5.3 Small-sample confidence intervals for μ

Example 5.5.8 The scores of a random sample of 16 people who took the TOEFL (Test of English as a Foreign Language) had a mean of 540 and a standard deviation of 50. Construct a 95% CI for the population mean μ of the TOEFL score, assuming that the scores are normally distribute

5.5.3 Small-sample confidence intervals for μ

Example 5.5.8 The scores of a random sample of 16 people who took the TOEFL (Test of English as a Foreign Language) had a mean of 540 and a standard deviation of 50. Construct a 95% CI for the population mean μ of the TOEFL score, assuming that the scores are normally distribute

Solution

Because $n = 16$ is small, using Theorem 5.5.1 with degrees of freedom 15, a 95% CI for μ is:

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} = 540 \pm 2.131 \left(\frac{50}{\sqrt{16}} \right)$$

So the 95% for the population mean μ of the TOEFL scores is $(513.36, 566.64)$.

5.5.3 Small-sample confidence intervals for μ

Example 5.5.9 The following data represent the total ozone levels measured in Dobson units at randomly selected locations on the Earth on a particular day:

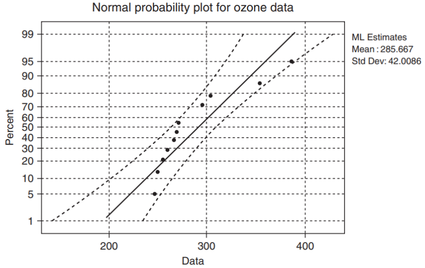
269	246	388	354	266	303
295	259	274	249	271	254

Example 5.5.9 The following data represent the total ozone levels measured in Dobson units at randomly selected locations on the Earth on a particular day:

269	246	388	354	266	303
295	259	274	249	271	254

Solution

The following is the probability plot of these data created using Minitab.



5.5.3 Small-sample confidence intervals for μ

Because all the data values lie within the bounds on the normal probability plot (see the discussion in Section 3.2.4), we can assume that the data have approximate normality. We have $\bar{x} = 285.7$ and $s = 43.9$. Also, $n = 12$. For $\alpha = 0.05$,$t_{0.025,11} = 2.201$. A 95% CI for μ is:

$$\bar{x} \pm t_{\alpha/2,(n-1)} \frac{s}{\sqrt{n}} = 285.7 \pm 2.201 \left(\frac{43.9}{\sqrt{12}} \right).$$

Hence, a 95% CI for μ, the average ozone level over the Earth, lies in $(257.81, 313.59)$.

Contents

5.6 A confidence interval for the population variance

Theorem 5.6.1 If $\bar{X}$ and S are the mean and standard deviation of a random sample of size n from a normal population, then:

$$P\left(\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2},n-1}} \le \sigma^2 \le \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2},n-1}}\right) = 1-\alpha$$

Example 5.6.1 A random sample of size 21 from a normal population gave a standard deviation of 9. Determine a 90% CI for σ^2.

5.6 A confidence interval for the population variance

Example 5.6.1 A random sample of size 21 from a normal population gave a standard deviation of 9. Determine a 90% CI for σ^2.

Solution

Here $n = 21$ and $s^2 = 81$. From the χ^2 table with 20 degrees of freedom, $\chi^2_{0.05} = 31.4104$ and $\chi^2_{0.95} = 10.8508$. Therefore, a 90% CI for σ^2 is obtained from:

$$\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}} \right)$$

Thus we get,

$$\frac{(20)(81)}{31.4104} < \sigma^2 < \frac{(20)(81)}{10.8508}$$

or we are 90% confident that $51.575 < \sigma^2 < 149.298$.

5.6 A confidence interval for the population variance

Assumption: The population is normal, we can get the procedure to find confidence interval for σ^2

- Calculate $\bar{x}$ and s^2 from the sample $x_1, \dots, x_n$
- Find $\chi^2_U = \chi^2_{\frac{\alpha}{2}, n-1}$ and $\chi^2_L = \chi^2_{1-\frac{\alpha}{2}, n-1}$ using the χ^2 square table with $(n-1)$ degrees of freedom
- Compute the $(1-\alpha)100\%$ CI for the population variance σ^2 as

$$\left(\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2}, n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}} \right)$$

Example 5.6.2

The following data represent cholesterol levels (in mg/dL) of 10 randomly selected patients from a large hospital on a particular day:

360	352	294	160	146	142	318	200	142	116

Determine a 95% CI for σ^2.

Example 5.6.2

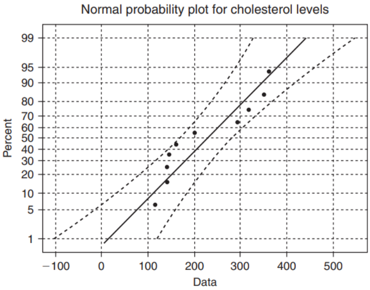
The following data represent cholesterol levels (in mg/dL) of 10 randomly selected patients from a large hospital on a particular day:

360	352	294	160	146	142	318	200	142	116

Determine a 95% CI for σ^2.

Solution

$\bar{x} = 223$ and $s = 96.9$. The following probability graph is obtained



5.6 A confidence interval for the population variance

We can assume approximate normality for the data. A box plot of the data shows that there are no outliers. $\chi^2_{0.025,9} = 19.023$ and $\chi^2_{0.975,9} = 2.70$. Therefore a 90% CI for σ^2 is obtained from:

$$\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}} \right)$$

Thus, we get:

$$\frac{(9)(96.9)^2}{19.023} < \sigma^2 < \frac{(9)(96.9)^2}{2.70}$$

or we are 95% confident that $4442.3 < \sigma^2 < 31,299$. Note that the numbers look very large, but it is the value of variance. By taking the square root of the numbers on the both sides, we can also get a CI for the standard deviation σ. As remarked in the previous exercise, in general, to find $a(1-\alpha)100\%$ CI for the true population standard deviation, σ, take the square roots of the end points of the CI of the variance.

Contents

5.7 CI concerning two population parameters

- We consider the interval estimation based on samples from two populations.
- Our aim is to obtain a CI for the parameters of interest based on two independent samples taken from these two populations

5.7 CI concerning two population parameters

- Let $X_{11}, \dots, X_{1n_1}$ be a random sample from a normal distribution with mean μ_1 and variance σ_1^2
- Let $X_{21}, \dots, X_{2n_2}$ be a random sample from a normal distribution with mean μ_2 and variance σ_2^2
- Let $\bar{X}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} X_{1i}$ and $\bar{X}_2 = \frac{1}{n_2}\sum_{i=1}^{n_2} X_{2i}$
- Assume that the two samples are independent
- $$\bar{X}_1 - \bar{X}_2 \sim N(\mu_1 - \mu_2, \frac{1}{n_1}\sigma_1^2 + \frac{1}{n_2}\sigma_2^2)$$

Assume that the population is normal, and the samples are independent. The large-sample confidence interval for the difference of two means:

- σ_1,σ_2 are known. The $(1-\alpha)100\%$ large sample CI for $\mu_1 - \mu_2$ is given by:

$$(\bar{X}_1 - \bar{X}_2) \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

- If σ_1 and σ_2 are not known, σ_1 and σ_2 can be replaced by the respective sample standard deviations S_1 and S_2 when $n_i \geq 30, i = 1, 2$. Thus, we can write:

$$P\left((\bar{X}_1 - \bar{X}_2) - z_{\frac{\alpha}{2}} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq (\bar{X}_1 - \bar{X}_2) + z_{\frac{\alpha}{2}} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}\right)$$

$$= 1 - \alpha$$

5.7 CI concerning two population parameters

Example 5.7.1 A study of two kinds of machine failures shows that 58 failures of the first kind took an average of 79.7 minutes to repair with a standard deviation of 18.4 minutes, whereas 71 failures of the second kind took on average 87.3 minutes to repair with a standard deviation of 19.5 minutes. Find a 99% CI for the difference between the true average amounts of time it takes to repair failures of the two kinds of machines.

Example 5.7.1 A study of two kinds of machine failures shows that 58 failures of the first kind took an average of 79.7 minutes to repair with a standard deviation of 18.4 minutes, whereas 71 failures of the second kind took on average 87.3 minutes to repair with a standard deviation of 19.5 minutes. Find a 99% CI for the difference between the true average amounts of time it takes to repair failures of the two kinds of machines.

Solution

Here, $n_1 = 58, n_2 = 71, \bar{x_1} = 79.7, s_1 = 18.4, \bar{x_2} = 87.3,$ and $s_2 = 19.5$. The the 99% CI for $\mu_1 - \mu_2$ is given by:

$$(79.7 - 87.3) \pm 2.575\sqrt{\frac{(18.4)^2}{58} + \frac{(19.5)^2}{71}}$$

- We are 99% certain that $\mu_1 - \mu_2$ is located in the interval $(-16.215, 1.0149)$.
- Note that$-16.215 < \mu_1 - \mu_2 < 1.0149$ means that more than 90% of the length of this interval is negative.
- We can conclude that μ_2 dominates μ_1, that is, $\mu_2 > \mu_1$ more than 90% of the time.

The small-sample case

- In the small-sample case, the problem of constructing CIs for the difference of the means from the two normal populations with unknown variances can be a difficult one
- If we assume that the two populations have a common but unknown variance, say $\sigma_1^2 = \sigma_2^2 = \sigma^2$, we can obtain an estimate of the variance by pooling the two sample data sets
- Assumption: The samples are independent from two normal populations with equal variances

5.7 CI concerning two population parameters

- Define the pooled sample variance S_p^2 as:

$$S_p^2 = \frac{\sum_{i=1}^{n_1}(X_{1i} - \bar{X}_1)^2 + \sum_{i=1}^{n_2}(X_{2i} - \bar{X}_2)^2}{n_1 + n_2 - 2} = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

- When the two samples are independent,

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

has a t distribution with $n_1 + n_2 - 2$ degrees of freedom

- Small-sample confidence interval for the difference of two means ($\sigma_1^2 = \sigma_2^2$)

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\frac{\alpha}{2},(n_1+n_2-2)} S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Example 5.7.2 Independent random samples from two normal populations with equal variances produced the following data:

Sample 1:	1.2	3.1	1.7	2.8	3
Sample 2:	4.2	2.7	3.6	3.9	

(a) Calculate the pooled estimate of σ^2.

(b) Obtain a 90% CI for $\mu_1 - \mu_2$.

Example 5.7.2 Independent random samples from two normal populations with equal variances produced the following data:

Sample 1:	1.2	3.1	1.7	2.8	3
Sample 2:	4.2	2.7	3.6	3.9	

(a) Calculate the pooled estimate of σ^2.

(b) Obtain a 90% CI for $\mu_1 - \mu_2$.

Solution
(a) We have $n_1 = 5$ and $n_2 = 4$. Also,

$$\bar{x}_1 = 2.36, s_1^2 = 0.733$$

$$\bar{x}_2 = 3.6, s_2^2 = 0.42.$$

Hence, $$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = 0.5989$$

(b) For the confidence coefficient 0.90,$\alpha = 0.10$, and from the t table, $t_{0.05,7} = 1.895$. Thus, a 90% CI for $\mu_1 - \mu_2$ is:

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2(n_1+n_2-2)} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$= (2.36 - 3.6) \pm 1.895\sqrt{0.5989(\frac{1}{5} + \frac{1}{4})}$$
$$= -1.24 \pm 0.98 = (-2.22, -0.26)$$

Here, μ_2 dominates μ_1 uniformly. Note that we can decrease the confidence range, -2.22 to -0.26, by increasing n_1 and n_2 with $1-\alpha = 0.90$ to remain the same. This means that we are closing on the unknown true value of $\mu_1 - \mu_2$.

- In the small-sample case, if the equality of the variances cannot be reasonably assumed, that is, $\sigma_1^2 \neq \sigma_2^2$; we can still use the previous procedure, except that we use the following degrees of freedom in obtaining the t value from the table. Let

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{(n_1 - 1)} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{(n_2 - 1)}}$$

- The number given in this formula is always rounded down for the degrees of freedom. Hence, in this case, a small sample $(1-\alpha)100\%$ CI for $\mu_1 - \mu_2$ is given by:

$$(\bar{X_1} - \bar{X_2}) \pm t_{\frac{\alpha}{2},\nu}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

  where the t distribution has ν degrees of freedom as given previously

Example 5.7.3 Assume that two populations are normally distributed with unknown and unequal variances. Two independent samples are taken with the following summary statistics:

$$\begin{array}{lll} n_1 = 16 & \bar{x}_1 = 20.17 & s_1 = 4.3 \\ n_2 = 11 & \bar{x}_2 = 19.23 & s_2 = 3.8 \end{array}$$

Construct a 95% CI for $\mu_1 - \mu_2$.

Solution

First let us compute the degrees of freedom,

$$v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}} = \frac{\left(\frac{(4.3)^2}{16} + \frac{(3.8)^2}{11}\right)}{\frac{\left(\frac{(4.3)^2}{16}\right)^2}{15} + \frac{\left(\frac{(3.8)^2}{11}\right)^2}{110}} = 23.312$$

Hence, $v = 23$, and $t_{0.025,23} = 2.069$.
Now a 95% CI for $\mu_1 - \mu_2$ is :

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,v}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = (20.17 - 19.23)$$

$$\pm(2.069)\sqrt{\frac{(4.3)^2}{16} + \frac{(3.8)^2}{11}}$$

which gives the 95% CI as:

$$-2.3106 < \mu_1 - \mu_2 < 4.1906$$

Assume that the two populations are normal, and the samples are independent. A $(1-\alpha)100\%$ confidence interval for $\frac{\sigma_1^2}{\sigma_2^2}$

$$\left(\left(\frac{S_1^2}{S_2^2}\right)\left(\frac{1}{F_{n_1-1,n_2-1,1-\frac{\alpha}{2}}}\right), \left(\frac{S_1^2}{S_2^2}\right)\left(\frac{1}{F_{n_1-1,n_2-1,\frac{\alpha}{2}}}\right) \right)$$

That is,

$$P\left(\left(\frac{S_1^2}{S_2^2}\right)\left(\frac{1}{F_{n_1-1,n_2-1,1-\frac{\alpha}{2}}}\right) \le \frac{\sigma_1^2}{\sigma_2^2} \le \left(\frac{S_1^2}{S_2^2}\right)\left(\frac{1}{F_{n_1-1,n_2-1,\frac{\alpha}{2}}}\right) \right)$$

$$= 1-\alpha$$

Example 5.7.5 Assuming that two populations are normally distributed, two independent random samples are taken with the following summary statistics:

$$\begin{aligned} n_1 = 21 \quad \bar{x}_1 = 20.17 \quad s_1 = 4.3 \\ n_2 = 16 \quad \bar{x}_2 = 19.23 \quad s_2 = 3.8 \end{aligned}$$

Example 5.7.5 Assuming that two populations are normally distributed, two independent random samples are taken with the following summary statistics:

$$\begin{array}{lll} n_1 = 21 & \bar{x}_1 = 20.17 & s_1 = 4.3 \\ n_2 = 16 & \bar{x}_2 = 19.23 & s_2 = 3.8 \end{array}$$

Solution

Here, $n_1 = 21, n_2 = 16$, and $\alpha = 0.05$. Using the F table, we have:

$$F_{n_1-1,n_2-1,1-\alpha/2} = F(20, 15, 0.975) = 2.76$$

and

$$F_{n_2-1,n_1-1,1-\alpha/2} = F(15, 20, 0.975) = 2.57$$

5.7 CI concerning two population parameters

A 95% CI for σ_1^2/σ_2^2 is:

$$\left(\left(\frac{S_1^2}{S_2^2}\right)\left(\frac{1}{F_{n_1-1,n_2-2,1-\alpha/2}}\right),\left(\frac{S_1^2}{S_2^2}\right)F_{n_2-1,n_1-1,1-\alpha/2}\right)$$

$$=\left(\left(\frac{(4.3)^2}{(3.8)^2}\right)\left(\frac{1}{2.76}\right),\left(\frac{(4.3)^2}{(3.8)^2}\right)(2.57)\right)=(0.46394,3.2908)$$

That is, we are 95% confident that the ratio of true variance, $\frac{\sigma_1^2}{\sigma_2^2}$, is located in the interval that implies a 95% $CI(0.46394,3.2908)$.

Contents

5.8 Chapter summary

- The basic concepts of estimation, both point estimation and interval estimation
- Two methods of finding point estimators:
  - The method of moments
  - The method of maximum likelihood
- Unbiasedness and MSE
- A $(1-\alpha)100\%$ CI for an unknown parameter θ is computed from sample data. The so-called pivotal method is introduced for deriving a CI.

Contents

5.9 Computer examples

- 5.9.1 Examples using R
  - Example 5.9.1 Descriptive point estimates
  - Example 5.9.2 Uniform maximum likelihood
  - Example 5.9.4 Confidence interval
  - Example 5.9.5 Confidence interval
- 5.9.2 Minitab examples
  Example 5.9.6-Example 5.9.10
- 5.9.3 SPSS examples
- 5.9.4 SAS examples

- Example 5.9.1 **Descriptive point estimates**
  Generate 50 sample points from an $N(4, 4)$ distribution and find the descriptive statistics. Obtain an unbiased and sufficient estimate of μ.

- Solution
  R-code

```
sample = rnorm(50,4,4)
summary(sample);
sd(sample);
sd(sample)/sqrt(length(sample));
```

  Output
  Your output will be unique since the samples are generated randomly; take notice of standard error.

Min.	1st Qu.	Median	Mean 3rd	Qu.	Max.
-4.292	1.105	4.012	3.865	6.478	14.790

  Notice "Mean 3rd" this is an estimate; we know that the population mean is 4 as we defined it. 4.288085 is standard deviation and

- Example 5.9.2 **Uniform maximum likelihood**
  Generate 35 samples from a $U(0,5)$ distribution and, using the descriptive statistics command, find the maximum likelihood estimate for these data.
  We know that for a random sample $X_1,\dots,X_n$ from $U(0,\theta)$, the MLE$\hat{\theta} = \max(X_i) = X_{(n)}$, the nth order statistic. We can use the following steps to obtain the estimate.
  R-code
  ```
  sample=runif(35,0,5);
  summary(sample);
  ```
  Output
  Your output will be unique since the samples are generated randomly

Min.	1st Qu.	Median	Mean 3rd	Qu.	Max.
0.1155	1.5710	2.9520	2.7620	4.0920	4.9900

- Example 5.9.4 **Confidence interval**

  For the following data obtain a 98% CI for μ:

Sample (x):								
	6.8	5.6	8.5	8.5	8.4	7.5	9.3	9.4
	7.8	7.1	9.9	9.6	9.0	13.7	9.4	16.6
	9.1	10.1	10.6	11.1	8.9	11.7	12.8	11.5
	10.6	12.0	11.1	6.4	12.3	12.3	11.4	9.9
	15.5	14.3	11.5	13.3	11.8	12.8	13.7	13.9
	12.9	14.2	14.0					

  This example assumes you have stored the data into variable x. Please modify your code appropriately.

  R-code t.test(x,conf.level=0.98);

  Output

  One Sample t-test data: x

  t = 27.7762, df = 42, p-value $<$ 2.2e-16

  alternative hypothesis: true mean is not equal to 0

  98 percent confidence interval: 9.910598 11.801030

  sample estimates: mean of x 10.85581

- Example 5.9.5 **Confidence interval**

  For the following data, find a 90% CI for $\mu_1 - \mu_2$ using the following data:

Sample (x):	1.2	3.1	1.7	2.8	3.0
Sample (y):	4.2	2.7	3.6	3.9	

  This example assumes you have stored your data into variables x and y. Please modify your code appropriately.

  R-code

  t.test(x,y,conf.level=0.90);

  Output

  Welch Two Sample t-test data: x and y

  t = -2.4721, df = 6.996, p-value = 0.04272

  alternative hypothesis: true difference in means is not equal to 0 90 percent confidence interval: -2.1903896 -0.2896104

  sample estimates:

mean of x	mean of y
2.36	3.60

Contents

5.10 Projects for Chapter 5

- 5.10.1 Asymptotic properties
- 5.10.2 Robust estimation
- 5.10.3 Numerical unbiasedness and consistency
- 5.10.4 Averaged squared errors
- 5.10.5 Alternate method of estimating the mean and variance
- 5.10.6 NewtoneRaphson in one dimension

5.10 Projects for Chapter 5

- 5.10.7 The empirical distribution function
- 5.10.8 Simulation of coverage of the small confidence intervals for μ
- 5.10.9 Confidence intervals based on sampling distributions
- 5.10.10 Large-sample confidence intervals: general case
- 5.10.11 Prediction interval for an observation from a normal population
- 5.10.12 Empirical distribution function as estimator for cumulative distribution function

Exercises and Projects

- Exercises
  - Exercises 5.2.3, 5.2.7; 5.2.10, 5.2.14, 5.2.15
  - Exercises 5.3.7, 5.3.10
  - Exercises 5.4.9
  - Exercises 5.5.4, 5.5.13, 5.2.28
  - Exercises 5.6.4
- Projects
  - 5.10.9 Confidence intervals based on sampling distributions